

# DIGITAL LOGIC & DESIGN (EE-227)

## ASSIGNMENT #1

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ID: \_\_\_\_\_

NAME: \_\_\_\_\_

SECTION: \_\_\_\_\_

### Read the Instructions Carefully

- ❖ Base 3, base 4, base 5, base 6 means the one you used in QUESTION 1
- ❖ Your assigned number is given to you in excel sheet provided with assignment

**FOR EXAMPLE:** Assigned Number if your assigned number is **0821**

|                                         | Assign Digit 0 | Assign Digit 1 | Assign Digit 2 | Assign Digit 3 |
|-----------------------------------------|----------------|----------------|----------------|----------------|
| Short for Assigned Digit                | A0             | A1             | A2             | A3             |
| Write Assigned Number<br>Digit By Digit | 7              | 8              | 2              | 1              |

**FOR EXAMPLE:** Name is **HAMAZADAUD**

- ❖ Use your name instead of HAMZA
- ❖ If your name starts with **MUHAMMAD** kindly use your second name
- ❖ Convert repeated character to small letter or to other symbols to make them unique (see example for **A**, **a** & **@**)

|                                              | ZERO<br>CHARACTER<br>OF YOUR<br>NAME | FIRST<br>CHARACTER<br>OF YOUR<br>NAME | SECOND<br>CHARACTER<br>OF YOUR<br>NAME | THIRD<br>CHARACTER<br>OF YOUR<br>NAME | FOURTH<br>CHARACTER<br>OF YOUR<br>NAME | FIFTH<br>CHARACTER<br>OF YOUR<br>NAME | SIXTH<br>CHARACTER<br>OF YOUR<br>NAME |
|----------------------------------------------|--------------------------------------|---------------------------------------|----------------------------------------|---------------------------------------|----------------------------------------|---------------------------------------|---------------------------------------|
| Short for<br>CHARACTER                       | C0                                   | C1                                    | C2                                     | C3                                    | C4                                     | C5                                    | C6                                    |
| YOUR<br>NAME<br>CHARACTER<br>BY<br>CHARACTER | H                                    | A                                     | M                                      | Z                                     | a                                      | D                                     | @                                     |

1. Fill the table given below and Count in given base

**NOTE:** Use Assigned number & Your Name

- **Base 3** (Zero Digit is Assign0 (**A0**), First Digit is Assign1 (**A1**), Second Digit is Assign2 (**A2**))
- **Base 4** (Zero Digit is Assign0 (**A0**), First Digit is Assign1 (**A1**), Second Digit is Assign2 (**A2**), and Third Digit is Assign3 (**A3**))
- **Base 5** (Zero Digit is Character0 (**C0**) of your name, First Digit is Character1 (**C1**) of your name, Second Digit is Character2 (**C2**) of your name, Third Digit is Character3 (**C3**) of your name, Four Digit is Character04 (**C4**) of your name)
- **Base 6** (Zero Digit is Character0 (**C0**) of your name, First Digit is Character1 (**C1**) of your name, Second Digit is Character2 (**C2**) of your name, Third Digit is Character3 (**C3**) of your name, Four Digit is Character04 (**C4**) of your name and Fifth Digit is Character05 (**C5**) of your name)

TABLE

| BASE 10<br>DECIMAL | BASE 2<br>BINARY | BASE 3 | BASE 4 | BASE 5 | BASE 6 | BASE 8<br>OCTAL | BASE 16<br>HEXA |
|--------------------|------------------|--------|--------|--------|--------|-----------------|-----------------|
| 0                  | 0                |        |        |        |        |                 |                 |
| 1                  | 1                |        |        |        |        |                 |                 |
| 2                  |                  |        |        |        |        |                 |                 |
| 3                  |                  |        |        |        |        |                 |                 |
| 4                  |                  |        |        |        |        |                 |                 |
| 5                  |                  |        |        |        |        |                 |                 |
| 6                  |                  |        |        |        |        |                 |                 |
| 7                  |                  |        |        |        |        |                 |                 |
| 8                  |                  |        |        |        |        |                 |                 |
| 9                  |                  |        |        |        |        |                 |                 |
| 10                 |                  |        |        |        |        |                 |                 |
| 11                 |                  |        |        |        |        |                 |                 |
| 12                 |                  |        |        |        |        |                 |                 |
| 13                 |                  |        |        |        |        |                 |                 |
| 14                 |                  |        |        |        |        |                 |                 |
| 15                 |                  |        |        |        |        |                 |                 |
| 16                 |                  |        |        |        |        |                 |                 |
| 17                 |                  |        |        |        |        |                 |                 |
| 18                 |                  |        |        |        |        |                 |                 |
| 19                 |                  |        |        |        |        |                 |                 |
| 20                 |                  |        |        |        |        |                 |                 |

ID:

ASSIGNMENT # 1 [EE-227]

Assigned Number:

| BASE 10 | BASE 2<br>BINARY | BASE 3 | BASE 4 | BASE 5 | BASE 6 | BASE 8<br>OCTAL | BASE 16<br>HEXA |
|---------|------------------|--------|--------|--------|--------|-----------------|-----------------|
| 21      |                  |        |        |        |        |                 |                 |
| 22      |                  |        |        |        |        |                 |                 |
| 23      |                  |        |        |        |        |                 |                 |
| 24      |                  |        |        |        |        |                 |                 |
| 25      |                  |        |        |        |        |                 |                 |
| 26      |                  |        |        |        |        |                 |                 |
| 27      |                  |        |        |        |        |                 |                 |
| 28      |                  |        |        |        |        |                 |                 |
| 29      |                  |        |        |        |        |                 |                 |
| 30      |                  |        |        |        |        |                 |                 |
| 31      |                  |        |        |        |        |                 |                 |
| 32      |                  |        |        |        |        |                 |                 |
| 33      |                  |        |        |        |        |                 |                 |
| 34      |                  |        |        |        |        |                 |                 |
| 35      |                  |        |        |        |        |                 |                 |
| 36      |                  |        |        |        |        |                 |                 |
| 37      |                  |        |        |        |        |                 |                 |
| 38      |                  |        |        |        |        |                 |                 |
| 39      |                  |        |        |        |        |                 |                 |
| 40      |                  |        |        |        |        |                 |                 |
| 41      |                  |        |        |        |        |                 |                 |
| 42      |                  |        |        |        |        |                 |                 |
| 43      |                  |        |        |        |        |                 |                 |
| 44      |                  |        |        |        |        |                 |                 |
| 45      |                  |        |        |        |        |                 |                 |
| 46      |                  |        |        |        |        |                 |                 |
| 47      |                  |        |        |        |        |                 |                 |
| 48      |                  |        |        |        |        |                 |                 |
| 49      |                  |        |        |        |        |                 |                 |
| 50      |                  |        |        |        |        |                 |                 |

2. Perform the following conversion to check your counting in question 1

•  $(25)_{10} = ( ? )_3$      Base 3 used in Question # 1

•  $(32)_{10} = ( ? )_4$      Base 4 used in Question # 1

•  $(41)_{10} = ( ? )_5$      Base 5 used in Question # 1

•  $(49)_{10} = ( ? )_6$      Base 6 used in Question # 1

3. Convert Decimal (Assigned Number)<sub>10</sub> to ( ? )<sub>3</sub>, ( ? )<sub>4</sub>, ( ? )<sub>5</sub> and ( ? )<sub>6</sub> check results by recovering them to decimal
4. Convert Decimal fraction (0. Last two digits of the Assigned Number)<sub>10</sub> to ( ? )<sub>3</sub>, ( ? )<sub>4</sub>, ( ? )<sub>5</sub> and ( ? )<sub>6</sub> check results by reconverting them to decimal
5. Perform addition to result of Question 3 to base in Question 1 as follow
  - (Assigned number)<sub>3</sub> + ( question 1 number 46)<sub>3</sub>
  - (Assigned number)<sub>4</sub> + ( question 1 number 49)<sub>4</sub>
  - (Assigned number)<sub>5</sub> + ( question 1 number 35)<sub>5</sub>
  - (Assigned number)<sub>6</sub> + ( question 1 number 24)<sub>6</sub>
6. Perform subtraction to result of Question 3 from base in Question 1 as follow
  - (Assigned number)<sub>3</sub> - ( question 1 number 46)<sub>3</sub>
  - (Assigned number)<sub>4</sub> - ( question 1 number 49)<sub>4</sub>
  - (Assigned number)<sub>5</sub> - ( question 1 number 35)<sub>5</sub>
  - (Assigned number)<sub>6</sub> - ( question 1 number 50)<sub>6</sub>
7. Perform R's and (R-1)'s complement to the results of question 3 to generate negative number
  - R's and (R-1)'s of (Assigned number)<sub>3</sub> of results from Q3
  - R's and (R-1)'s of (Assigned number)<sub>4</sub> of results from Q3
  - R's and (R-1)'s of (Assigned number)<sub>5</sub> of results from Q3
  - R's and (R-1)'s of (Assigned number)<sub>6</sub> of results from Q3

8. Perform R-1 's complement subtraction using values from Question 7 and Question 1
- (question 1 number 15)<sub>3</sub> - (R-1)'s of (Assigned number)<sub>3</sub> results from Q7
  - (question 1 number 25)<sub>4</sub> - (R-1)'s of (Assigned number)<sub>4</sub> results from Q7
  - (question 1 number 33)<sub>5</sub> - (R-1)'s of (Assigned number)<sub>5</sub> results from Q7
  - (question 1 number 37)<sub>6</sub> - (R-1)'s of (Assigned number)<sub>6</sub> results from Q7
9. Perform R 's complement subtraction using values from Question 7 and Question 1
- (question 1 number 13)<sub>3</sub> - (R)'s of (Assigned number)<sub>3</sub> results from Q7
  - (question 1 number 17)<sub>4</sub> - (R)'s of (Assigned number)<sub>4</sub> results from Q7
  - (question 1 number 23)<sub>5</sub> - (R)'s of (Assigned number)<sub>5</sub> results from Q7
  - (question 1 number 44 )<sub>6</sub> - (R)'s of (Assigned number)<sub>6</sub> results from Q7
10. Convert you assigned number to BCD. Add with 1369 using BCD addition
11. Convert you assigned number to BCD. Subtract 999 from assigned number using BCD subtraction
12. Convert you assigned number to (?)<sub>2</sub> and generate next 3 Gray codes.
13. Consider gray codes generated in Q12 and convert them back to binary
14. Number Multiplication
- Multiply Last two digits of your assigned number(**A2A3**)<sub>10</sub> with (**17**)<sub>10</sub>
  - Multiply (**15**)<sub>3</sub> x (**5**)<sub>3</sub> =(**?**)<sub>3</sub> Result should be in base 3 as well
  - Convert assigned Number **A3** to binary and replace in given number (**A3.011**)<sub>2</sub> with (**101.011**)<sub>2</sub>
15. Number Division
- Divide First two digits of your assigned number(**A0A1**)<sub>10</sub> with (**3**)<sub>10</sub>
  - Convert assigned Number (**A0A1**)<sub>10</sub> to binary and replace in given number (**A0A1**)<sub>2</sub> with (**11**)<sub>2</sub>